

Background

This work presents part of a PhD research project aiming to design accessible representations for people who are **blind** or have **low vision (BLV)** in understanding body movement. The first phase of the research involved understanding key challenges of body movement education from both the BLV people's perspective and the **teachers'** perspective. We conducted 2 surveys, 23 interviews and 4 focus groups [1]. The current phase of the research involves exploring potential solutions and **designing** artefacts. Two ideation workshops were conducted with researchers representing different human centered computing disciplines in particular data visualization and inclusive technologies. We are currently exploring the feasibility of the **design space** and planning further ideation workshops to be conducted with BLV people and dance teachers.

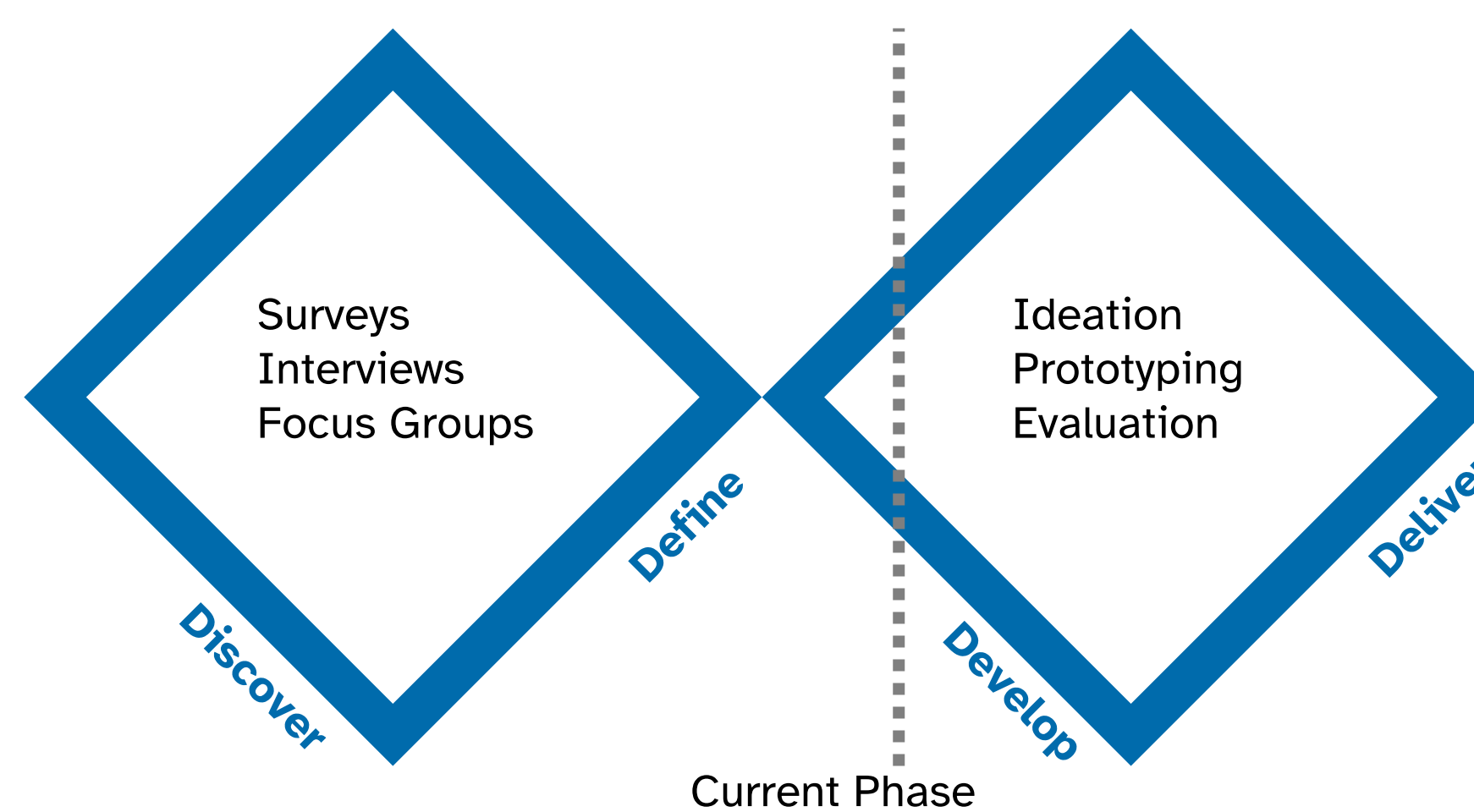


Figure 1. Overall Methodology: Double Diamond Framework. (Design Council's "Double Diamond")



Figure 2. Ideation Workshops with Researchers

Key Design Challenge

How can the body movement elements be represented effectively using different **modalities** for understanding and experiencing dance by BLV people?

"never being able to sort of watch dance, I just didn't even have an idea about some of the ways your body could move, or the ways like that you could move your arms through space..."
~ a BLV participant

We present the ongoing exploration of the design space for representing dance movement **elements** defined by Laban framework [2] through different modalities. It is explored under synchronous (in situation and real time) and asynchronously (pre or post-movement). Table 1 is an overview of our preliminary ranking of how those elements can be represented effectively using different modalities based on 1) insights shared by participants regarding their familiarity with the tools and techniques, 2) previous insights from related work, and 3) an initial exploration conducted by ourselves.

Table 1. Design space and applications for different modalities. A preliminary ranking of 1 (low) to 3 (high) represents the effectiveness and feasibility of each. ? indicates further investigation is needed.

Modality and Tools	Body	Directions	Levels	Multiple dancers	Time	Force
Tactile: Refreshable Displays	2	2	3	2	2	1
Tactile: 3D Models	3	3	2	?	1	?
Auditory: Sonification	?	2	2	1	3	2
Auditory: Music Mixing	?	2	2	1	3	2
Haptics: Wearables	1	2	2	?	2	1

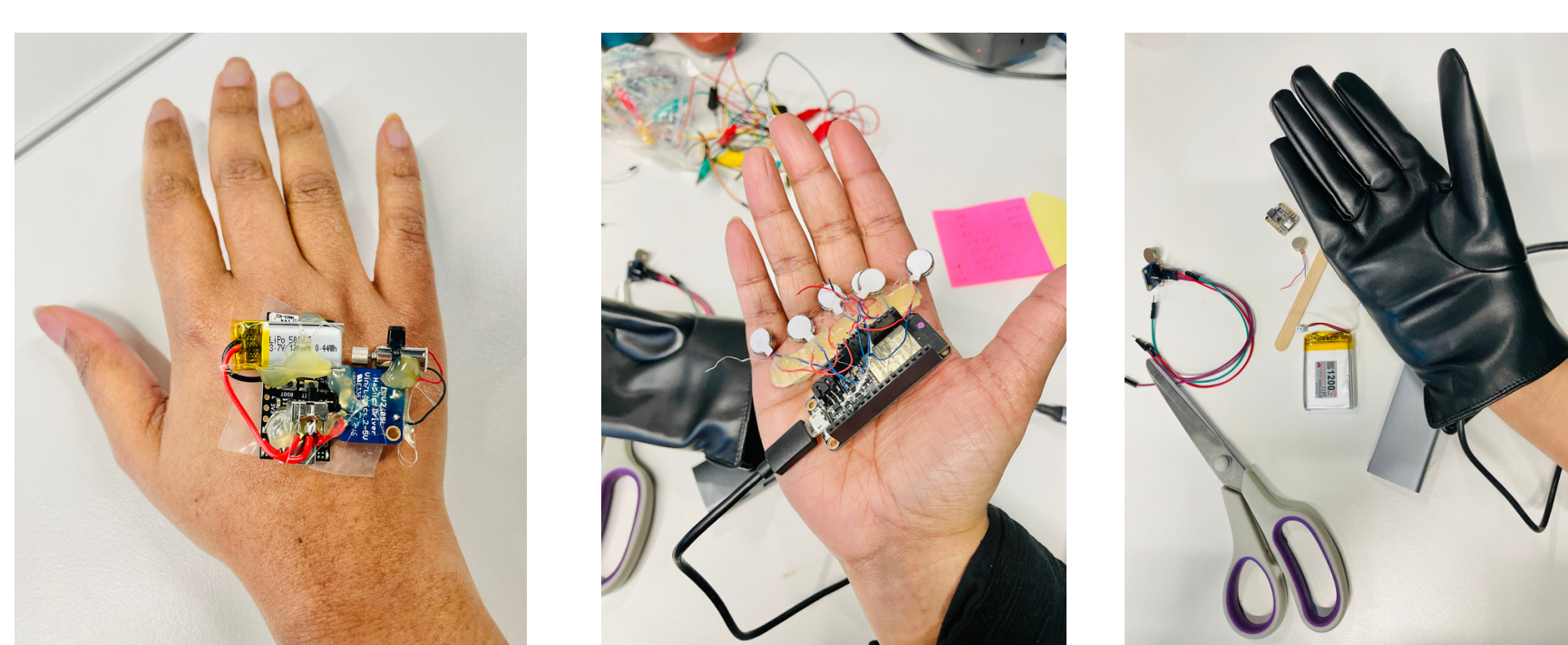
Sonification and Music Mixing

Sonification have been explored by some dance-based studies [5] mapping movement to auditory qualities of sound (intensity, frequency, loudness). We use a similar approach expanding to music mixing, e.g., using isolated individual tracks within a song, placed spatially around the dancer, and real-time, **controllable effects** applied to these.



Haptics and Somaesthetics

Haptic feedback-based solutions have been explored to understand whole-body navigation and particular limb movement but less research conducted on fine-tuning the quality of movement. The somatic pedagogy and trending somaesthetics [6] can facilitate **sensory experiences** of body movement.



Refreshable Tactile Displays

Refreshable tactile displays (RTDs) with multiple heights are superior to 2.5D tactile graphics for showing relative body positions and beneficial in showing a sequence of diagrams for progressive motion. This aligns with **data physicalization** concepts using tactile variables such as roughness [3].

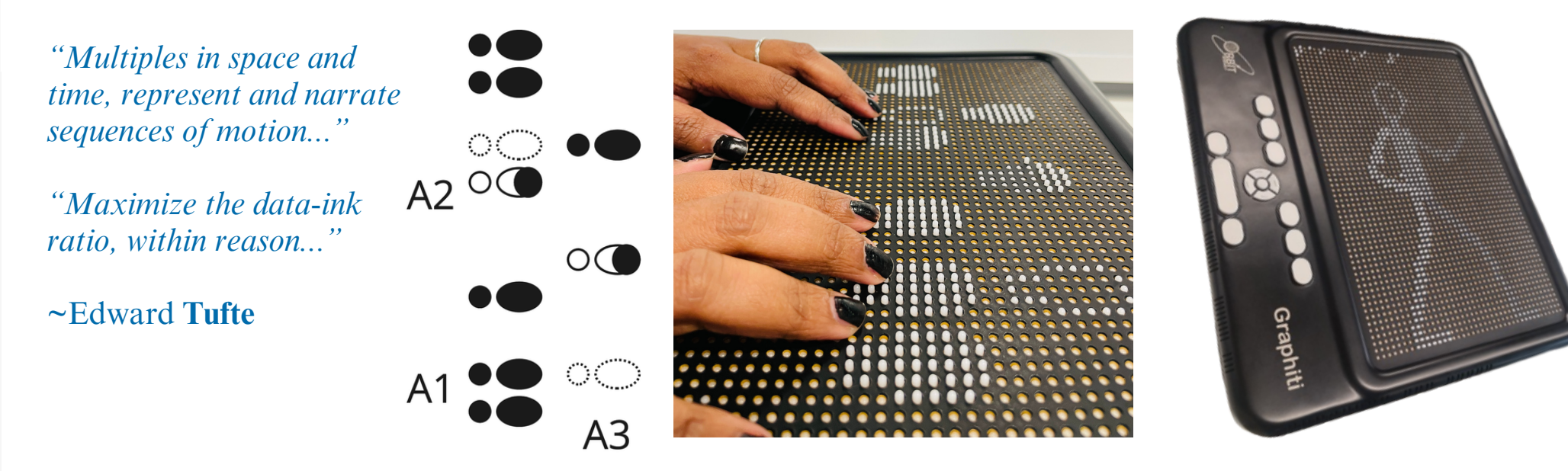


Figure 4. Graphiti: a refreshable tactile display

3D Models

Studies show that 3D models can be easier to understand than 2.5D tactile graphics, especially in spatial understanding for blind people [4]. Moveable tactile models or robots could potentially be used to show body positions and movements when there are issues with person-to-person **body touch** or when learning remotely.



Figure 5. Manipulative 2D and 3D Figures

Figure 3. The movement of teacher is tracked using Motion Capture (MoCap) cameras and represented through sound (music tracks played through headphones) and vibrations (glove with vibration motors).

Contact Information

Madhuka De Silva
Department of Human Centred Computing
Monash University
madhuka.desilva@monash.edu
Twitter.com/@madhuka_designs



References

- [1] Madhuka De Silva, et al. Understanding Challenges and Opportunities in Body Movement Education of People who are Blind or have Low Vision. ASSETS '23, USA (in press).
- [2] R. Laban and L. Ullmann. The mastery of movement. Dance Books Ltd., Hampshire, UK, 2011.
- [3] S. Stusak, et al. Variables for data physicalization units. IEEE VIS, 2018.
- [4] L. Holloway, et al. Accessible maps for the blind: Comparing 3d printed models with tactile graphics. CHI '18, USA.
- [5] J. R. Dias, et al. Tele-media-art: Feasibility tests of web-based dance education for the blind using kinect and sound synthesis of motion. Int. J. Technol. Hum. Interact., 2019.
- [6] Kristina Höök, et al. 2021. Unpacking Non-Dualistic Design: The Soma Design Case. ACM Trans. Comput.-Hum. Interact.

